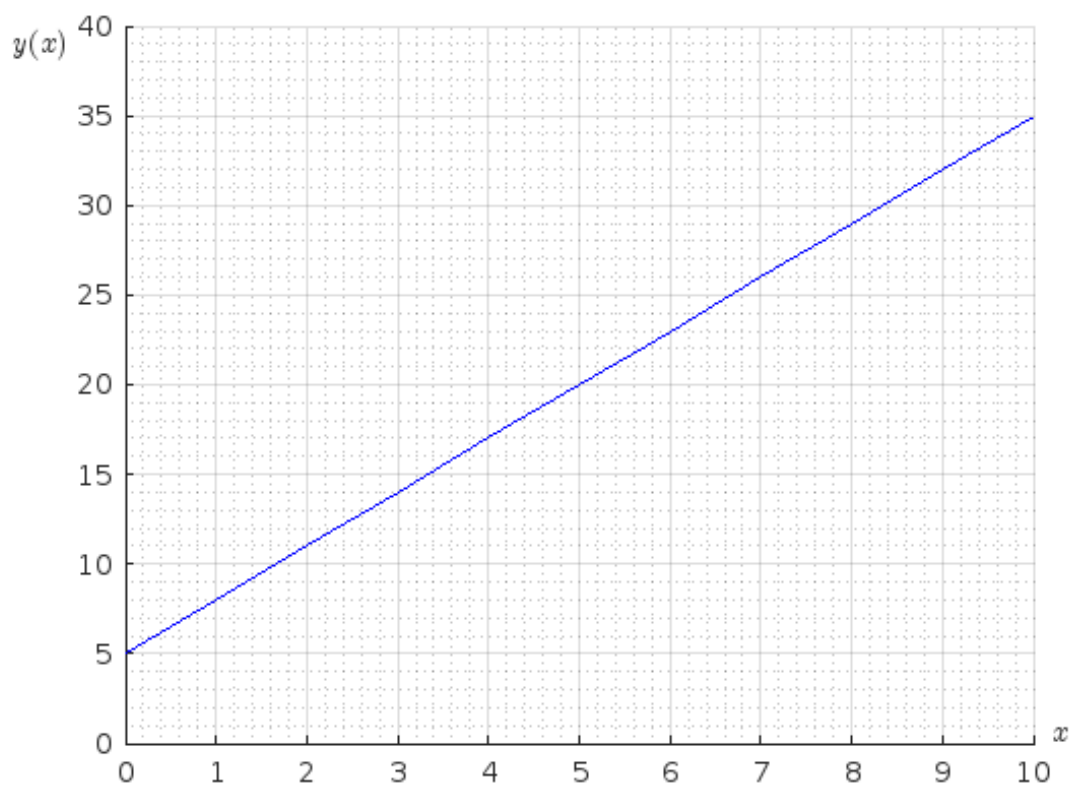


Rail Engineering Mathematics

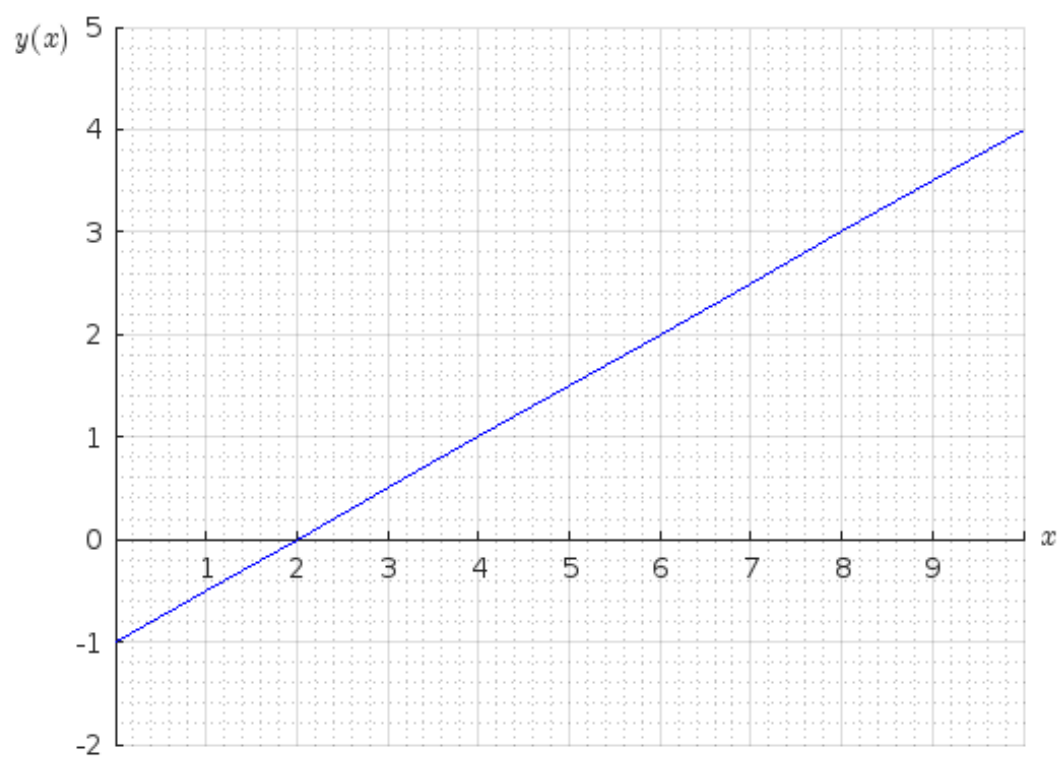
Tutorial Sheet 1

1. Determine the equation of the line from the following graphs:

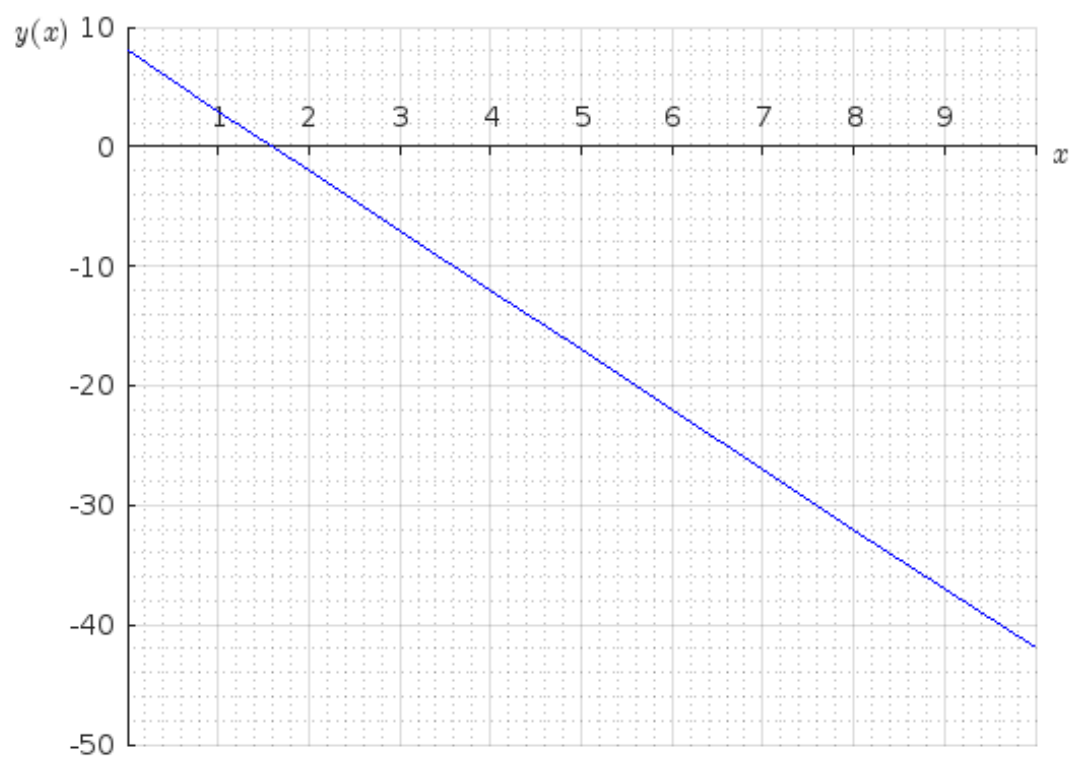
(a)



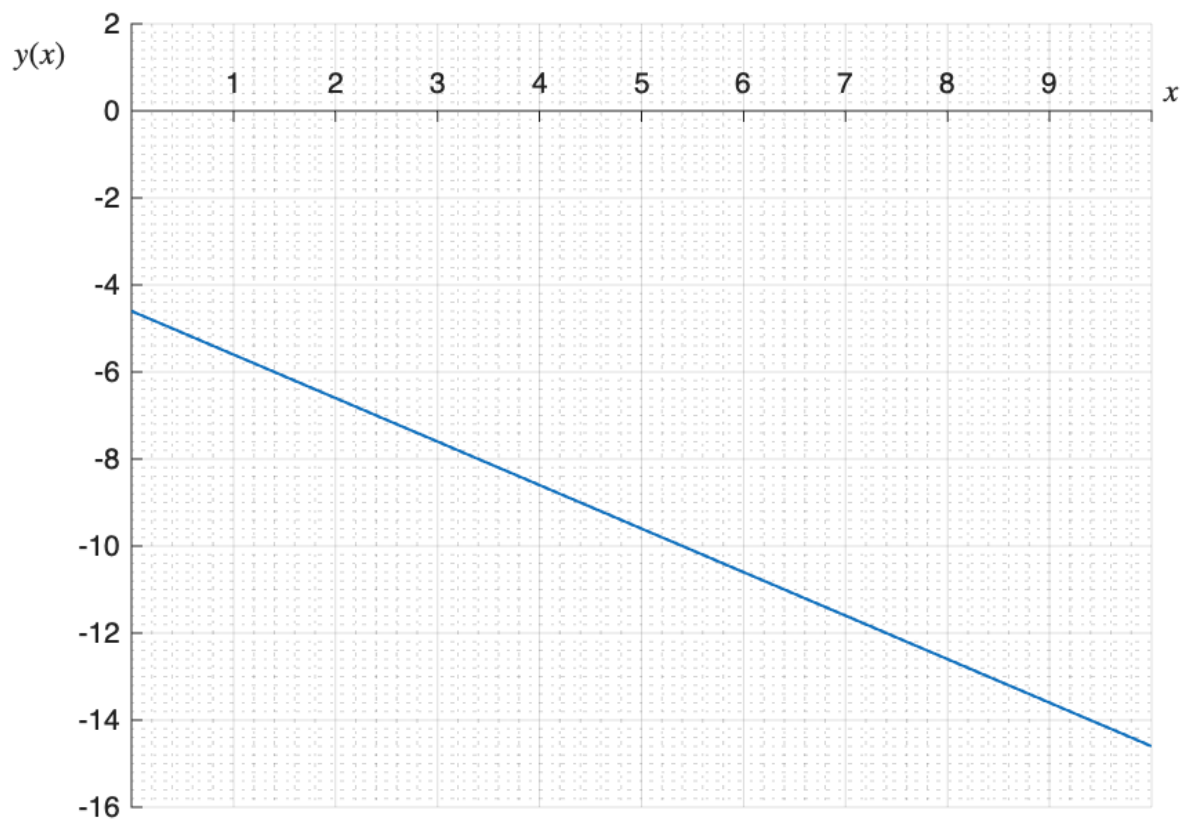
(b)



(c)



(d)



2. Sketch the graph of the following functions, indicating the points at which the line crosses the x and y axes:

(a) $y = 3x + 2$

(b) $y = 3x - 2$

(c) $y = 3x$

(d) $y = 1 - 2x$

(e) $y = -x + 1$

(f) $y = 1$

(g) $y = 1 + 2x$

3. Using technology (MATLAB or otherwise), plot the following polynomial functions. You may choose your own range of x , unless otherwise stated.

(a) $y = 3x - 9$

(b) $y = -5x + 12$

(c) $y = 7x^2 - 4x + 6$ in the range $-1 \leq x \leq 3$

(d) $y = x^3 - 6x^2 + 8x - 1$ in the range $-1 \leq x \leq 5$

(e) $y = -2x + 6$ and $y = x^2 - 6x - 16$ in the range $-4 \leq x \leq 10$. Plot both of these functions on the same set of axes.

4. Determine the equation of the straight line that passes through the points:

(a) $(-2, 9)$ and $(1, -6)$

(b) $(-1, -8)$ and $(3, 4)$

(c) $(3, -13)$ and $(5, -17)$

(d) $(1, 5.5)$ and $(6, 3)$

5. You have been tasked with analysing the behaviour of a complex component in a circuit. You have a dataset of current (I) and voltage (V) measurements, but it is not immediately obvious if the relationship is linear. The data points are as follows:

I (amps)	V (volts)
1	3
2	5
3	8
4	11
5	15

First, determine whether the relationship between current (I) and voltage (V) is linear. If it is, find the linear equation that describes this relationship. Then, predict the voltage (V) when the current is 6 A. How might understanding this behaviour be critical for designing and optimizing circuits in practical engineering applications?

6. In a market, when there is high demand for a product compared to its supply (before the market gets saturated with product A or a similar one from a competitor), we can roughly estimate the relationship between the profit of the company and the number of units manufactured using a linear model.

You are a financial analyst in the employ of the **Persimmon** company, which has just developed a new product called the **uPhone**. Public reception has been extremely positive, and market research suggests that most people are very keen to purchase a **uPhone** as soon as possible. Currently the supply of **uPhones** coming out of the factories is relatively low due to the complex engineering processes and rare metals required in the manufacturing process.

In the first quarter of this year, the company made and sold 1500 units, and reported a profit of £60,000. In the second quarter, the company were able to increase production to 6000 units. All of these were sold and a profit of £960,000 was made.

(a) How many **uPhones** need to be sold each quarter in order to break even during that financial quarter?

(b) In the third quarter, supply managers are warning that production may slow to 5000 units. Assuming that public demand for the product remains high, how much profit do you forecast?

7. Consider the quadratic function $y = x^2 + 3x - 4$

(a) Where does $y = x^2 + 3x - 4$ cross the y -axis?

(b) Where does $y = x^2 + 3x - 4$ cross the x -axis?

(c) Is it $\cup$ -shaped, or $\cap$ -shaped?

(d) Sketch the graph of this function.

8. Repeat the process of Question 7 for the functions:

(i) $y = 16 - x^2$

(ii) $y = 2x^2 + 4x - 20$

Check your results by plotting the curves in MATLAB or other software (such as EXCEL, Desmos or GeoGebra).

9. Calculate the roots of the following polynomials:

(a) $y = x^2 + 6x - 27$

(b) $y = x^2 + 2x - 48$

(c) $y = 3x^2 + 11x + 8$

(d) $y = 6x^2 - 23x + 20$

10. If the roots of the quadratic $y = ax^2 + bx + c$ are x_1 and x_2 then it is possible to rewrite the quadratic in its factorised form as follows:

$$y = a(x - x_1)(x - x_2)$$

- (a) Use this information to rewrite the four quadratics that appear in Question 9.

- (b) Expand the brackets of your new form in each case to demonstrate that it is equivalent to the original form.

11. Solve $5x^2 + 2x = 4$, giving the solutions correct to 3 significant figures.

12. Solve $x^2 = 4(x - 3)^2$

13. Solve $\frac{5}{x+2} + \frac{9}{x-2} = 2$

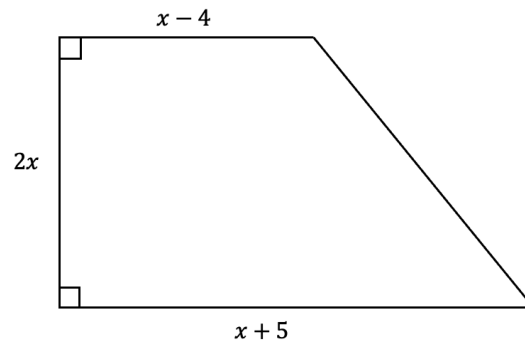
14. The quadratic formula is being used to solve a quadratic equation. After substitution of the coefficients into the formula, we have

$$x = \frac{-7 \pm \sqrt{17}}{4}$$

Work out the original quadratic equation, giving your answer in the form $ax^2 + bx + c = 0$.

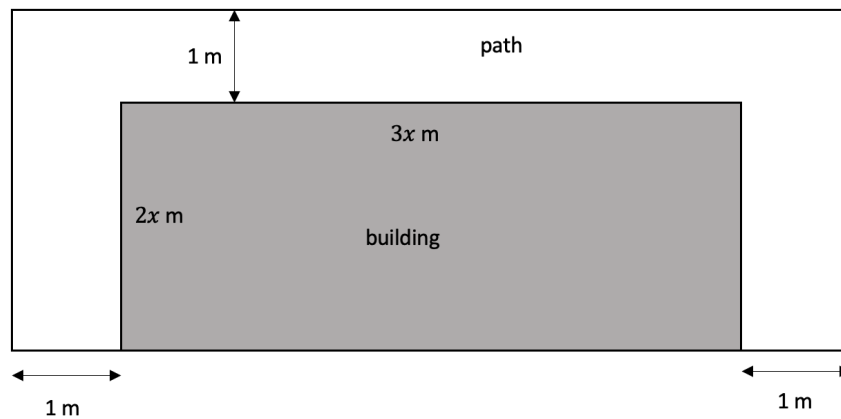
15. Solve the equation $\frac{2^{(n^2)}}{2^n \times 2^6} = 1$

16. Shown in the diagram below is a trapezium (not drawn to scale) where all measurements are in centimetres.



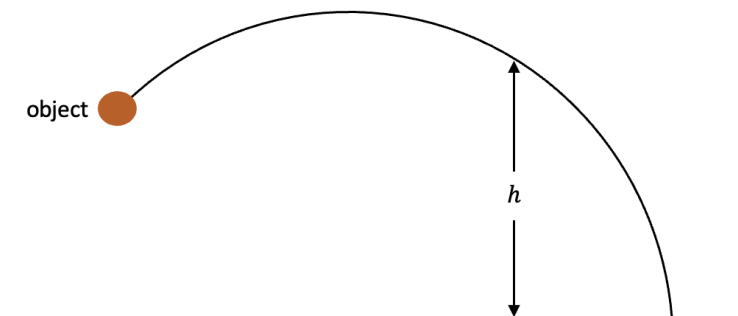
The area of the trapezium is 351 m^2 . Determine the value of x .

17. A rectangular building has a length of $3x$ metres and a width of $2x$ metres. The building has a path of width 1 metre on three of its sides, as depicted in the diagram below. Note that the diagram is not drawn to scale.



Given that the total area of the building and path is 100 m^2 calculate the area of the building.

18. An object is being launched from an initial height of 2 m above the ground. The object follows a parabola in the form $h = 2 + 6t - 5t^2$, where h is the height in metres above the ground t seconds after it has been launched.



The object hits the ground after T seconds. Calculate the value of T , giving the answer to 2 decimal places.

19. Access the **Interactive Function Explorer** on the module Blackboard site. Navigate to the “logarithmic” tab, featuring the general logarithmic equation:

$$y = A \ln(Bx) + C$$

Determine how the two parameters affect the shape of the graph:

(a) A

(b) C

What happens when you try to calculate $\ln(0)$ or $\log$ of a negative number?

20. Access the **Interactive Function Explorer** on the module Blackboard site. Navigate to the “exponential” tab, featuring the general exponential function:

$$y = A e^{Bx} + C$$

Determine how the two parameters affect the shape of the graph:

(a) A

(a) B

(b) C

What conditions must be satisfied by B in order to obtain exponential growth, or to obtain exponential decay?

21. Solve the following exponential equations for x :

(a) $e^x = 9.5$

(b) $5e^x = 50.6$

(c) $27.3e^x - 12 = 112.7$

(d) $-14.6e^x + 7.5 = 4.2$

(e) $29 - 6.3e^x = 15.4$

(f) $34.7e^{4.2x} + 4.3 = 25.1$

(g) $117.1e^{-0.4x} - 15.7 = -3.8$

22. Solve the following logarithmic equations for x :

(a) $\ln(x) = 0.6$

(b) $15 - \ln(2x) = 10.3$

(c) $2.8 = \log_{10}(0.3x + 6)$

23. The pressure p pascals at height h metres above ground level is given by:

$$p = p_0 e^{-\frac{h}{C}}$$

where p_0 is the pressure at ground level and C is a constant.

Find the pressure p when $p_0 = 1.012 \times 10^5$ Pa, height is $h = 1420$ m and $C = 71500$.

24. The current i flowing in a capacitor at time t is given by:

$$i = 12.5 \left(1 - e^{-\frac{t}{CR}} \right)$$

where $R = 30$ kilohms, and the capacitance C is 20 micro-farads.

Determine: (a) the current flowing after 0.5 seconds. (b) the time for the current to reach 10 amperes.

25. The resistance R of an electrical conductor at temperature $\theta^\circ C$ is given by $R = R_0 e^{\alpha\theta}$, where α is a constant and $R_0 = 5 \times 10^3$ ohms.

Determine the value of α , correct to 4 significant figures, when $R = 6 \times 10^3$ ohms and $\theta = 1500^\circ C$.

Also, find the temperature, correct to the nearest degree, when the resistance R is 5.4×10^3 ohms.

26. A research group models the population P (in millions) of a particular country by following formula:

$$P = Ae^{kt},$$

where t is the time (in years) since 1980, the value of k is 0.0241, and A is a constant that is yet to be determined.

In the year 2000, the population of the country was recorded as 11 million.

- (a) What is the population projected to be in 2020?
- (b) When is the population forecast to exceed 25 million?
- (c) Plot the projected population in MATLAB between the years 1980 and 2050. From the graph, describe the behaviour of this exponential function.
- (d) From the graph, determine when the population is equal to 27 million. Indicate clearly on the graph how you obtain your solution.
- (e) Your colleague uses this formula to predict the population of the country in the year 2500. Discuss the reliability of this prediction.

27. The temperatures θ_1 and θ_2 of a pipe with inner radius r_1 and outer radius r_2 are given by:

$$\theta_1 = -\frac{Q}{2\pi kL} \ln(r_1) \quad \text{and} \quad \theta_2 = -\frac{Q}{2\pi kL} \ln(r_2)$$

where Q is the heat transfer rate, L is the length of the pipe and k is the thermal conductivity. Show that:

$$Q = \frac{2\pi kL(\theta_1 - \theta_2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

28. The voltage v and current i of an inductor is given by:

$$i = 5(e^{-200t} - e^{-800t}) \quad \text{and} \quad v = e^{-200t} + 400e^{-800t}$$

Find an expression for the power $p = vi$ of the inductor.